Sahaj Bhadja

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Summary

Full-stack software engineer with 2+ years building **Python services, React applications, distributed data flows, and customer-facing analytics**, with experience in event-driven systems, experimentation, reliability, observability, and end-to-end product delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Retrieval, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Architected a multi-tenant platform with **normalized schemas, row-level isolation, indexed queries, and customer-facing analytics**, reducing dashboard latency from 1.2 seconds to 350 milliseconds for 500+ daily users.
- Restructured asynchronous FastAPI orchestration with **explicit job states, concurrent execution, failure isolation, monitoring, and production debugging**, tripling throughput for 1K+ jobs while maintaining 99%+ reliability across releases.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Owned React and Flask workflows for uploads, grading history, model controls, and feedback, supporting 150+ instructors and 10K+ daily requests through **rapid full-stack delivery, testing, and iteration**.
- Built concurrent Python ingestion for PDF, DOCX, ZIP, OCR, and multimodal data, reducing turnaround 60% through **secure validation, fault-tolerant execution, production troubleshooting, and repeatable CI/CD-backed production releases**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Designed cross-platform APIs and distributed data pipelines using **Python, AWS Glue, S3, SQL, and modular service interfaces**, reliably improving response times 30% while reducing technical debt 20%.
- Automated 20+ unit and integration suites with PyTest, Mockito, Jenkins, and virtual machines, strengthening **data consistency, regression coverage, observability, code review quality, and production issue investigation** across releases.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built React and Node.js workflows across eight services for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records through **REST APIs, WebSockets, event-driven updates, and shared schemas**.
- Automated Docker and Jenkins releases, reducing deployment time 50% while using **Sentry monitoring, Google Analytics, structured logs, regression testing, and product-performance data** to guide frequent reliable iterations.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built Flask APIs for bookings, pricing, coupons, payments, and real-time status updates across 50K+ records, applying **event tracking, reliable Firebase synchronization, webhook processing, and sub-200-millisecond response optimization**.
- Designed the end-to-end workflow for 12+ integrations and presented technical decisions to clients, translating **ambiguous business requirements into reliable scoped services, data flows, analytics, and implementation plans**.

Skills

Languages & Full Stack: Python, Java, JavaScript, TypeScript, React, Node.js, FastAPI, Flask, HTML5, CSS3
Distributed Systems: REST APIs, Microservices, Event-Driven Architecture, WebSockets, Background Jobs, Concurrency, Idempotency, Data Consistency, Fault Tolerance
Data & Analytics: PostgreSQL, MySQL, SQL Server, SQL, Redis, MongoDB, Data Modeling, Event Tracking, Attribution, Product Analytics, Experimentation
Cloud & Delivery: AWS, Azure, GCP, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Git, Linux
Quality & Product: PyTest, Mockito, Unit Testing, Integration Testing, Debugging, Observability, Monitoring, Code Reviews, Agile, Product Judgment, AI-Assisted Development

Projects

Promotion Auction and Delivery Platform | *Python, FastAPI, PostgreSQL, Redis, React, Docker*

- Designed a **promoted-content auction and delivery platform** that ranks eligible campaigns, enforces budgets and pacing, resolves competing bids, applies relevance signals, and exposes auditable delivery decisions through versioned APIs.

Attribution Analytics Pipeline | *Python, PostgreSQL, Redis, FastAPI, React, AWS*

- Built an **event-driven attribution pipeline** linking impressions, clicks, apply starts, and completed applications, with deduplication, delayed-event reconciliation, campaign reporting, experiment cohorts, and employer-facing analytics for trustworthy outcome measurement.

Trend Tracker with Fashion Recommendation System | *CNN, React, Material UI, Strapi, AWS EC2* | Live

- Developed a **recommendation-driven marketplace experience** using CNN models and React, improving recommendation accuracy from 78% to 92% while supporting personalized discovery across more than 3M users and 200K positive retentions.